College of Arts, Science & Commerce (Empowered Autonomous)

SIES

RISE WITH EDUCATION

NAAC REACCREDITED "A" GRADE

SIES College of Arts, Science and Commerce(Empowered Autonomous),

Sion (W), Mumbai – 400 022.

Department of Information Technology

CERTIFICATE

This is to certify that Mr. / ~~Miss.~~ MOSES JAIRAJ of BSc [Information Technology] HONOURS Semester VII, Seat No. HNSIT2627010 has successfully completed the practicals for the subject of RESEARCH METHODOLOGY as a partial fulfillment of the degree B.Sc(IT) HONOURS during the academic year 2026-27.

Faculty-in-Charge

Sudha Bhagavatheeswaran

Course Co-ordinator Date : External Examiner

Sudha Bhagavatheeswaran

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	ANNEXURES (attached as pdf files in the assignment in MS teams)
i)	Research papers – Literature survey
ii)	Questionnaire - Google Form Responses (excel sheet)

Practical 1:- Defining a research problem

- Problem definition - process of defining and developing a decision statement and the steps involved in translating it into more precise research terminology, including a set of research objectives.
- A decision statement is a written expression of the key question(s) that a research user wishes to answer.

Writing Managerial Decision Statements and Corresponding Research Objectives

Firm's Situation	Symptoms	Probable Problem	Decision Statement
Conduct Situation Analysis including interviews with key decision makers		Consider results of probing and apply creative processes	Express in actionable terms and make sure decision makers are in agreement
Manufacturer of palm-sized computer with wireless Internet access believes B2B sales are too low	• Distributors complain prices are too high • Business users still use larger computers	• Business users do not see advantages of smaller units • Advantages are not outweighed by costs • Transition costs may be a drawback for B2B customers more than for B2C customers	What things can be done to improve competitive positioning of the new product in B2B markets?

Decision Statement	Research Objectives	Research Questions	Research Hypotheses
Express in actionable terms and make sure decision makers are in agreement	Expresses potential research results that should aid decision-making	Ask a question that corresponds to each research objective	Specific statement explaining relationships, usually involving two variables, and including the direction of the relationship
What product features can be improved and emphasized to improve competitive positioning of the new product in B2B markets?	List actions that may overcome the objections (switching costs) of B2B customers toward adoption of the new product.	What are the factors that most lead to perceptions of high switching costs?	*Perceived difficulty* in learning how to use the new device is related to *switching costs*. *Price* is positively related to *switching costs*. *Knowledge* of new product is positively related to *switching costs*.

Decision Statement:-

To determine how the popularity and media coverage of cricket affect the growth and popularity of other sports in India.

Research Objective:-

To study the impact of cricket's popularity and media coverage on public interest in other sports in India.

Research Questions:-

1. How popular is cricket compared to other sports in India?
2. How does the popularity and media coverage of cricket influence the popularity of other sports in India?
3. What factors contribute to the dominance and popularity of cricket in India?
4. What factors affect the growth and promotion of other sports in India?
5. What measures can be taken to increase the popularity and growth of other sports in India?

Research Hypotheses:-

H_0 (Null Hypothesis): Cricket's popularity and media coverage have no significant impact on other sports in India.

H_1 (Alternative Hypothesis): Cricket's popularity and media coverage have a significant impact on other sports in India.

Practical 2:- Literature Review using search tools like google scholar

Google Scholar provides a simple way to broadly search for scholarly literature. Google Scholar helps you find relevant work across the world of scholarly research.

Google Scholar includes journal and conference papers, theses and dissertations, academic books, pre-prints, abstracts, technical reports and other scholarly literature from all broad areas of research. You'll find works from a wide variety of academic publishers, professional societies and university repositories, as well as scholarly articles available anywhere across the web. Google Scholar also includes court opinions and patents.

Features of Google Scholar

- Search all scholarly literature from one convenient place
- Explore related works, citations, authors, and publications
- Locate the complete document through your library or on the web
- Keep up with recent developments in any area of research
- Check who's citing your publications, create a public author profile

Finding recent papers

Your search results are normally sorted by relevance, not by date. To find newer articles, try the following options in the left sidebar:

1. click "Since Year" to show only recently published papers, sorted by relevance;
2. click "Sort by date" to show just the new additions, sorted by date;
3. click the envelope icon to have new results periodically delivered by email.

Locating the full text of an article

Abstracts are freely available for most of the articles. Alas, reading the entire article may require a subscription. Here're a few things to try:

1. click a library link, e.g., "FindIt@Harvard", to the right of the search result;
2. click a link labeled [PDF] to the right of the search result;
3. click "All versions" under the search result and check out the alternative sources;
4. click "Related articles" or "Cited by" under the search result to explore similar articles.

If you're affiliated with a university, but don't see links such as "FindIt@Harvard", please check with your local library about the best way to access their online subscriptions. You may need to do search from a computer on campus, or to configure your browser to use a library proxy.

Getting better answers

- If you're new to the subject, it may be helpful to pick up the terminology from secondary sources. E.g., a Wikipedia article for "overweight" might suggest a Scholar search for "pediatric hyperalimentation".
- If the search results are too specific for your needs, check out what they're citing in their "References" sections. Referenced works are often more general in nature.
- Similarly, if the search results are too basic for you, click "Cited by" to see newer papers that referenced them. These newer papers will often be more specific.
- Explore! There's rarely a single answer to a research question. Click "Related articles" or "Cited by" to see closely related work, or search for author's name and see what else they have written.

Google Scholar library

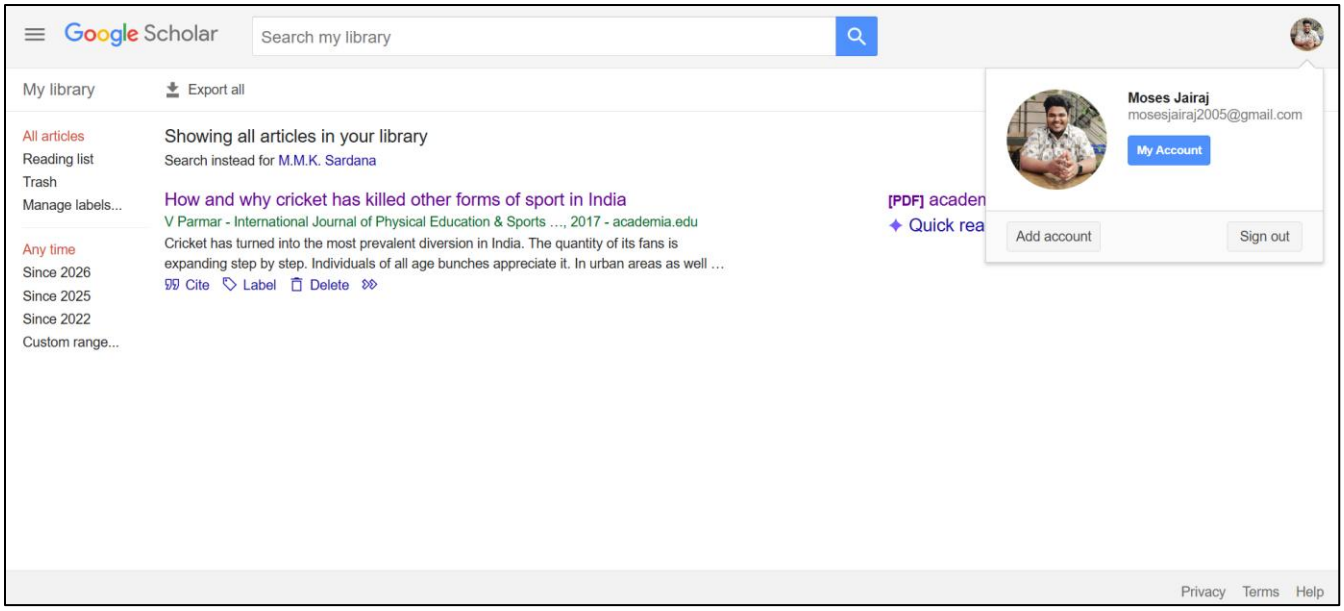
Google Scholar library is your personal collection of articles. You can save articles right off the search page, organize them by topic, and use the power of Scholar search to quickly find just the one you want - at any time and from anywhere. You decide what goes into your library, and we'll keep the links up to date.

You get all the goodies that come with Scholar search results - citing articles, related articles, formatted citations, links to your university's subscriptions, and more. Best of all, it's easy to quickly fill up your library with the articles you want - with a single click, you can import all the articles in your public Scholar profile as well as all the articles they cite.

Find the article you want to add in Google Scholar and click the "Save" link under the search result.

Click "My library" at the top of the Scholar homepage or in the left column of a search results page to view all articles in your library. To search the full text of these articles, enter your query as usual in the search box.

Snapshot of My library – Google Scholar



Details of research papers used for literature survey

1.

Topic:- Cricket Vs Other Sports: The Indian Scenario

By:- Dr. Vipendra Singh Parmar

Details of the study:-

The study examines the popularity and dominance of cricket in India and discusses how cricket's media coverage, sponsorship, infrastructure, prize money and popularity may affect the growth and visibility of other sports such as hockey, badminton, football, tennis and chess. The paper also discusses reasons behind cricket's popularity and possible measures to encourage other sports.

Data Collection Method:- Secondary data / Literature-based analysis using existing information and references.

Sample:- Cricket and other sports in India, including hockey, badminton, football, tennis and chess.

Sample Size:- Not specified.

2.

Topic:- Breaking the Cricket Barrier: Assessing the Reasons for the Underrepresentation of Other Sports in India

By:- Yash Khati, Pranauv S, Risabh Jain V, Chippy Mohan

Details of the study:-

The study examines the dominance of cricket in India and how it contributes to the underrepresentation of other sports. It discusses historical factors, media coverage, education, funding, infrastructure and socio-economic factors that influence the popularity and development of cricket compared with other sports

Data Collection Method:- Literature review / secondary research based on existing studies, reports and published sources.

Sample:- Not applicable, the study is based on existing literature and secondary sources rather than survey respondents.

Sample Size:- Not specified / Not applicable.

3.

Topic:- Sports in India Sans Cricket

By:- Prof. Sameer A. Virani and Dr. Srini R. Srinivasan

Details of the study:-

The study examines whether emerging sports and leagues such as football, tennis, ICL and Pro Kabaddi have affected the popularity and preference for cricket in India. It also compares the excitement associated with cricket, football and tennis.

Data Collection Method:- Primary data was collected through a survey/questionnaire. The responses were analysed using statistical methods such as ANOVA and a paired-samples test.

Sample:- Respondents from Mumbai, mainly in the age group of 18–24 years.

Sample Size:- 42 respondents.

Practical 3:- Research design

- "A research design is the arrangement of conditions for collection and analysis of data in a manner that aims to combine relevance to the research purpose with economy in procedure."
- Questions to be answered for research design

(i) What is the study about?

The study is about the impact of cricket's popularity on the growth, viewership, and public interest in other sports in India.

(ii) Why is the study being made?

The study is conducted to understand whether the popularity and media coverage of cricket influence people's interest in other sports.

(iii) What type of data is required?

Primary data is required, collected directly from respondents through a Google Form survey.

(iv) What will be the sample design?

The study uses 110 respondents selected through convenience sampling.

(v) What techniques of data collection will be used?

Data is collected using a structured questionnaire through Google Forms.

(vi) How will the data be analysed?

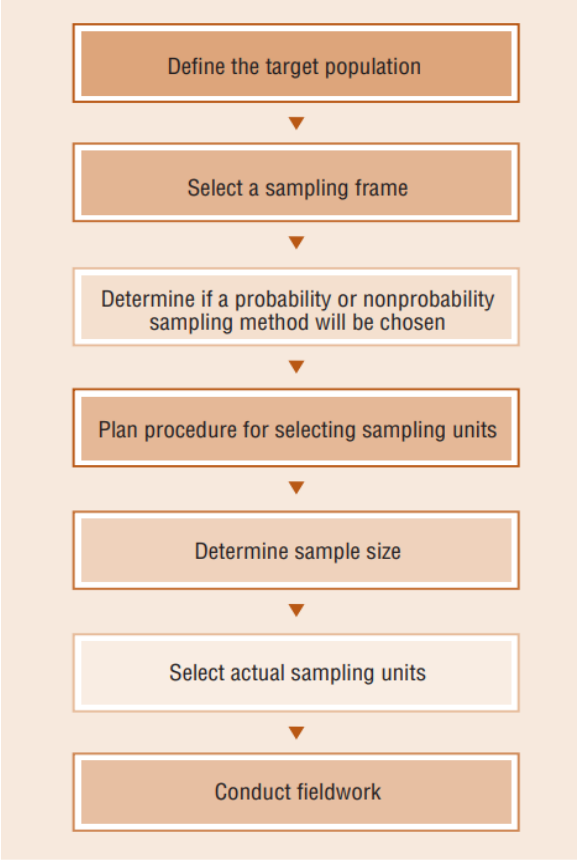
The collected data will be analysed using tabulation, graphs/charts, Chi-square test, t-test, ANOVA, and correlation analysis.

Practical 4:-Sampling Design

- The selected respondents constitute what is technically called a ‘sample’ and the selection process is called ‘sampling technique.’
- The survey so conducted is known as ‘sample survey’.
- The respondents selected should be as representative of the total population as possible in order to produce a miniature cross-section.

CHARACTERISTICS OF A GOOD SAMPLE DESIGN

- Sample design must result in a truly representative sample.
- Sample design must be such which results in a small sampling error.
- Sample design must be viable in the context of funds available for the research study.
- Sample design must be such so that systematic bias can be controlled in a better way.
- Sample should be such that the results of the sample study can be applied, in general, for the universe with a reasonable level of confidence.



Sample Characteristics:-

1. Target Population:- People in India who have an interest in or follow sports.
2. Sampling Method:- Non-Probability – Convenience Sampling
3. Sample Size:- 110 respondents

Practical 5:- Usage of measurement and scaling techniques

- Measurement is a process of mapping aspects of a domain onto other aspects of a range according to some rule of correspondence.
- In measuring, we devise some form of scale in the range (in terms of set theory, range may refer to some set) and then transform or map the properties of objects from the domain (in terms of set theory, domain may refer to some other set) onto this scale.

MEASUREMENT SCALES

- nominal scale
- ordinal scale
- ratio scale
- interval scale

Questions in the questionnaire	Measurement Scale
Q1. What is your age group?	**Ordinal**
Q2. Gender	**Nominal**
Q3. What is your occupation?	**Nominal**
Q4. Which sport do you follow the most?	**Nominal**
Q5. How often do you watch cricket?	**Ordinal**
Q6. How often do you watch sports other than cricket?	**Ordinal**
Q7. Which non-cricket sport do you enjoy watching the most?	**Nominal**
Q8. Have you ever attended a live match of a sport other than cricket?	**Nominal**
Q9. Where do you usually get sports-related news?	**Nominal**
Q10. Do you think cricket gets more media coverage than other sports?	**Nominal**
Q11. Do you think this more media coverage affects the popularity of other sports?	**Nominal**
Q12. Why do you think cricket is more popular than other sports?	**Nominal**
Q13. Which factor influences you the most to watch a sport?	**Nominal**

Q14. “The popularity of cricket limits the growth of other sports in India.”	**Interval**
Q15. “Other sports deserve the same level of promotion and media coverage as cricket.”	**Interval**
Q16. Would you watch other sports more if they were promoted like cricket?	**Ordinal**
Q17. Do you think social media has helped increase the popularity of sports other than cricket?	**Nominal**
Q18. Is the government doing enough to support other sports?	**Nominal**
Q19. Which TWO things would help other sports become more popular?	**Nominal**
Q20. Do you think other sports can become as popular as cricket in the future?	**Nominal**

Practical 6:-Testing of Hypothesis

- *Null hypothesis and alternative hypothesis*

A **null hypothesis** is a **hypothesis** that says there is no statistical significance between the two variables. It is usually the **hypothesis** a researcher or experimenter will try to disprove or discredit. An **alternative hypothesis** is one that states there is a statistically significant relationship between two variables.

The null hypothesis is generally symbolized as H_0 and the alternative hypothesis as H_a.

The null hypothesis and the alternative hypothesis are chosen before the sample is drawn

Alternative hypothesis	*To be read as follows*
$H_a : \mu \neq \mu_{H_0}$	(The alternative hypothesis is that the population mean is not equal to 100 i.e., it may be more or less than 100)
$H_a : \mu > \mu_{H_0}$	(The alternative hypothesis is that the population mean is greater than 100)
$H_a : \mu < \mu_{H_0}$	(The alternative hypothesis is that the population mean is less than 100)

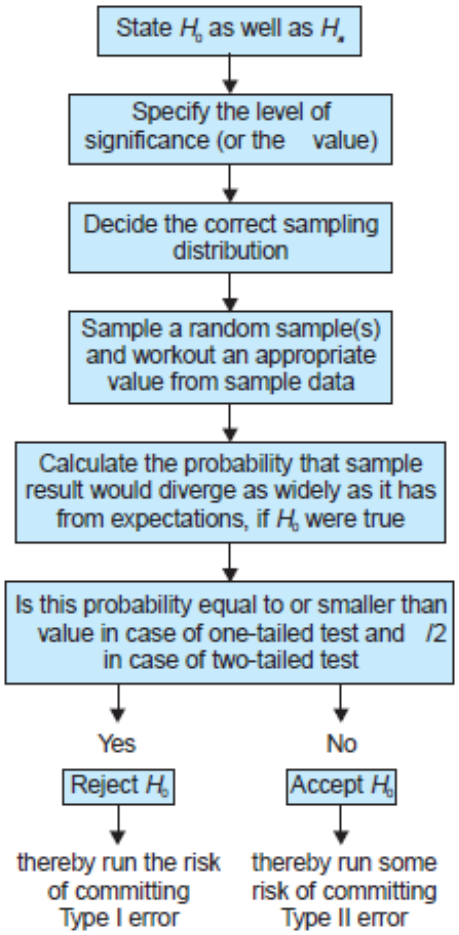
If we accept $H0$, then we are rejecting H_a and if we reject H_0, then we are accepting H_a.

- The level of significance

The significance level, also denoted as alpha or α, is the probability of rejecting the null hypothesis when it is true. For example, a significance level of 0.05 indicates a 5% risk of concluding that a difference exists when there is no actual difference.

- It is always some percentage (usually 5%) which should be chosen with great care, thought and reason.
- In case we take the significance level at 5 per cent, then this implies that H_0 will be rejected when the sampling result (i.e., observed evidence) has a less than 0.05 probability of occurring if H_0 is true.
- In other words, the 5 per cent level of significance means that researcher is willing to take as much as a 5 per cent risk of rejecting the null hypothesis when it ($H0$) happens to be true.
- Thus the significance level is the maximum value of the probability of rejecting H_0 when it is true and is usually determined in advance before testing the hypothesis.

FLOW DIAGRAM FOR HYPOTHESIS TESTING



Null Hypothesis:-

There is no significant difference or relationship between the popularity of cricket and the growth, popularity and promotion of other sports in India.

Alternative Hypothesis:-

There is a significant difference or relationship between the popularity of cricket and the growth, popularity and promotion of other sports in India.

Final Decision:-

To arrive at a final decision, Chi-Square Test, T-Test, ANOVA and Correlation Analysis were performed. The null hypothesis was rejected for the Chi-Square Test, T-Test and Correlation Analysis, while it was not rejected for the ANOVA test because its result was not statistically significant.

Practical 7:- Implement data analysis techniques

STATISTICS IN RESEARCH

- Measures of central tendency or statistical averages;

Mean; Arithmetic, Geometric, Harmonic

$$\text{Mean (or } \overline{X})^* = \frac{\sum X_i}{n} = \frac{X_1 + X_2 + \ldots + X_n}{n}$$

where $\overline{X}$ = The symbol we use for mean (pronounced as X bar)

$\sum$ = Symbol for summation

X_i = Value of the ith item X, $i = 1, 2, \ldots, n$

n = total number of items

In case of a frequency distribution, we can work out mean in this way:

$$\overline{X} = \frac{\sum f_i X_i}{\sum f_i} = \frac{f_1 X_1 + f_2 X_2 + \ldots + f_n X_n}{f_1 + f_2 + \ldots + f_n = n}$$

Sometimes, instead of calculating the simple mean, as stated above, we may workout the weighted mean for a realistic average. The weighted mean can be worked out as follows:

$$\overline{X}_w = \frac{\sum w_i X_i}{\sum w_i}$$

where $\overline{X}_w$ = Weighted item

w_i = weight of ith item X

X_i = value of the ith item X

Geometric mean is also useful under certain conditions. It is defined as the nth root of the product of the values of n times in a given series. Symbolically, we can put it thus:

$$\text{Geometric mean (or G.M.)} = \sqrt[n]{\pi X_i}$$

$$= \sqrt[n]{X_1 \cdot X_2 \cdot X_3 \ldots X_n}$$

where

G.M. = geometric mean,

n = number of items.

X_i = ith value of the variable X

π = conventional product notation

For instance, the geometric mean of the numbers, 4, 6, and 9 is worked out as

$$\text{G.M.} = \sqrt[3]{4.6.9}$$

$$= 6$$

The most frequently used application of this average is in the determination of average per cent of change i.e., it is often used in the preparation of index numbers or when we deal in ratios.

Harmonic mean is defined as the reciprocal of the average of reciprocals of the values of items of a series. Symbolically, we can express it as under:

$$\text{Harmonic mean (H.M.)} = \text{Rec.}\frac{\sum \text{Rec}X_i}{n}$$

$$= \text{Rec.}\frac{\text{Rec.}X_1 + \text{Rec.}X_2 + \ldots + \text{Rec.}X_n}{n}$$

where

H.M. = Harmonic mean
Rec. = Reciprocal
X_i = ith value of the variable X
n = number of items

For instance, the harmonic mean of the numbers 4, 5, and 10 is worked out as

$$\text{H.M.} = \text{Rec}\frac{1/4 + 1/5 + 1/10}{3} = \text{Rec}\frac{\frac{15+12+6}{60}}{3}$$

$$= \text{Rec}\left(\frac{33}{60} \times \frac{1}{3}\right) = \frac{60}{11} = 5.45$$

Median

$$\text{Median } (M) = \text{Value of}\left(\frac{n+1}{2}\right)\text{th item}$$

Median is a positional average and is used only in the context of qualitative phenomena, for example, in estimating intelligence, etc., which are often encountered in sociological fields. Median is not useful where items need to be assigned relative importance and weights. It is not frequently used in sampling statistics.

Mode

- *Mode* is the most commonly or frequently occurring value in a series.
- Like median, mode is a positional average and is not affected by the values of extreme items. it is, therefore, useful in all situations where we want to eliminate the effect of extreme variations.

- Measures of dispersion;

Range: *Range* is the simplest possible measure of dispersion and is defined as the difference between the values of the extreme items of a series. Thus,

The utility of range is that it gives an idea of the variability very quickly, but the drawback is that range is affected very greatly by fluctuations of sampling.

mean deviation,

$$\text{Mean deviation from mean } (\delta_{\bar{X}}) = \frac{\Sigma|X_i - \bar{X}|}{n}, \quad \text{if deviations, } |X_i - \bar{X}|, \text{ are obtained from arithmetic average.}$$

or

$$\text{Mean deviation from median } (\delta_m) = \frac{\Sigma|X_i - M|}{n}, \quad \text{if deviations, } |X_i - M|, \text{ are obtained from median}$$

or

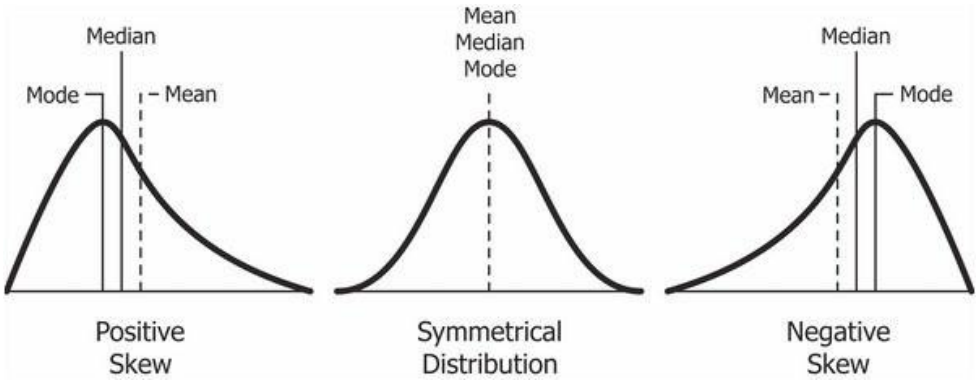
$$\text{Mean deviation from mode } (\delta_z) = \frac{\Sigma|X_i - Z|}{n}, \quad \text{if deviations, } |X_i - Z|, \text{ are obtained from mode.}$$

where δ = Symbol for mean deviation (pronounced as delta);
X_i = ith values of the variable X;
n = number of items;
$\bar{X}$ = Arithmetic average;
M = Median;
Z = Mode.

Standard deviation is most widely used measure of dispersion of a series.Standard deviation is defined as the square-root of the average of squares of deviations, when such deviations for the values of individual items in a series are obtained from the arithmetic average.It is popularly used in the context of estimation and testing of hypotheses

$$\text{Standard deviation}^* (\sigma) = \sqrt{\frac{\Sigma(X_i - \bar{X})^2}{n}}$$

- Measures of asymmetry (skewness)



- Measures of relationship

  Does there exist association or correlation between the two (or more) variables? If yes, of what degree? - CORRELATION

  Is there any cause and effect relationship between the two variables in case of the bivariate population or between one variable on one side and two or more variables on the other side in case of multivariate population? If yes, of what degree and in which direction? – REGRESSION

❖ TABULATION

Table 1: Age Group		
Response	**Frequency**	**Percentage**
18 - 25	98	89%
26 - 35	8	7%
Below 18	2	2%
Above 45	2	2%
Total	**110**	**100%**

Table 2: Gender		
Response	**Frequency**	**Percentage**
Female	58	53%
Male	51	46%
Prefer not to say	1	1%
Total	**110**	**100%**

Table 3: Occupation		
Response	**Frequency**	**Percentage**
Student	70	64%
Working Professinal	25	23%
Self-employed	8	7%
Homemaker	6	5%
Unemployed	1	1%
Total	**110**	**100%**

Table 4: Sport Followed Most		
Response	**Frequency**	**Percentage**
Cricket	63	57%
Kabaddi	12	11%
Badminton	11	10%
Football	11	10%
Tennis	3	3%
Volleyball	2	2%
Basketball	2	2%
Athletics	2	2%
Chess	1	1%
Both cricket and football	1	1%
Don't watch sports	1	1%
Boxing	1	1%
Total	**110**	**100%**

Table 5: Cricket Watching Frequency		
Response	**Frequency**	**Percentage**
Rarely	50	45%
Monthly	19	17%
Never	17	15%
Weekly	14	13%
Daily	10	9%
Total	**110**	**100%**

Table 6: Non-Cricket Sports Watching Frequency		
Response	**Frequency**	**Percentage**
Rarely	55	50%
Never	24	22%
Monthly	14	13%
Weekly	12	11%
Daily	5	5%
Total	**110**	**100%**

Table 7: Favourite Non-Cricket Sport		
Response	**Frequency**	**Percentage**
Kabaddi	38	35%
Football	37	34%
Badminton	10	9%
Volleyball	6	5%
Athletics	6	5%
Tennis	5	5%
No	1	1%
Null	1	1%
Golf	1	1%
Chess	1	1%
Wrestling	1	1%
Hockey	1	1%
Formula 1	1	1%
Total	**110**	**100%**

Table 8: Attended Live Non-Cricket Match		
Response	**Frequency**	**Percentage**
No	84	76%
Yes	26	24%
Total	**110**	**100%**

Table 9: Source of Sports News		
Response	**Frequency**	**Percentage**
Social Media	61	55%
Television	20	18%
Youtube	12	11%
Friends / Family	9	8%
Sports Apps / Websites	7	6%
Newspaper	1	1%
Total	**110**	**100%**

Table 10: Cricket Gets More Media Coverage		
Response	**Frequency**	**Percentage**
Yes	75	68%
Maybe	27	25%
No	8	7%
Total	**110**	**100%**

Table 11: Effect of Cricket Media Coverage on Other Sports		
Response	**Frequency**	**Percentage**
Yes	62	56%
Not Sure	34	31%
No	14	13%
Total	**110**	**100%**

Table 12: Why Cricket Is More Popular		
Response	**Frequency**	**Percentage**
Public interest	34	31%
Better media coverage	28	25%
Greater sponsorship	18	16%
Historical popularity	13	12%
Star players	12	11%
Better infrastructure	5	5%
Total	**110**	**100%**

Table 13: Main Factor Influencing Sport Viewership		
Response	**Frequency**	**Percentage**
Favorite player	35	32%
Exciting matches	33	30%
Friends/Family	22	20%
Major tournaments	11	10%
Media coverage	7	6%
Passion	1	1%
Total	**110**	**100%**

Table 14: Cricket Popularity Limits Growth of Other Sports (1–5)		
Response	**Frequency**	**Percentage**
1	8	7%
2	10	9%
3	28	25%
4	35	32%
5	29	26%
Total	**110**	**100%**

Table 15: Other Sports Deserve Equal Promotion/Media Coverage (1–5)		
Response	**Frequency**	**Percentage**
5	42	38%
4	34	31%
3	20	18%
2	7	6%
1	7	6%
Total	**110**	**100%**

Table 16: Would Watch Other Sports More if Promoted Like Cricket		
Response	**Frequency**	**Percentage**
Probably Yes	39	35%
Definitely Yes	32	29%
Not Sure	30	27%
Probably No	6	5%
Definitely No	3	3%
Total	**110**	**100%**

Table 17: Social Media Has Increased Popularity of Other Sports		
Response	**Frequency**	**Percentage**
Yes	73	66%
Maybe	28	25%
No	9	8%
Total	**110**	**100%**

Table 18: Government Doing Enough to Support Other Sports		
Response	**Frequency**	**Percentage**
Partially	34	31%
No	32	29%
Yes	22	20%
Not Sure	22	20%
Total	**110**	**100%**

Table 19: Two Things That Would Help Other Sports Become More Popular		
Response / Selected Option	**Frequency (Respondents)**	**Percentage of Respondents**
More media coverage	65	59%
Increased government funding	50	45%
Better sports infrastructure	41	37%
More international tournaments in India	35	32%
Better coaching facilities	25	23%
Grassroots sports programs in schools and colleges	19	17%
More sponsorship from private companies	18	16%
Celebrity endorsements	8	7%
Total respondents	**110**	

NOTE: This is a multiple-response question, so percentages may total more than 100% because respondents could select multiple options.

Table 20: Other Sports Can Become as Popular as Cricket		
Response	**Frequency**	**Percentage**
Yes	69	63%
Maybe	32	29%
No	9	8%
Total	**110**	**100%**

❖ ANALYSIS

1. T-Test – Q14

Question Selected

Q14: "The popularity of cricket limits the growth of other sports in India."

Why was Q14 selected?

Q14 was selected because it directly measures the main idea of the research, which is whether the popularity of cricket limits the growth of other sports in India.

The question uses a 1–5 Likert scale:

1 – Strongly Disagree
2 – Disagree
3 – Neutral
4 – Agree
5 – Strongly Agree

The test was used to check whether the average response of the respondents was significantly different from the neutral value of 3.

Steps Followed

1. The responses for Q14 from all 110 respondents were collected.
2. The responses were converted into numerical values from 1 to 5.
3. The mean and standard deviation of the responses were calculated.
4. The neutral value 3 was taken as the reference value.
5. The T-value was calculated using the One-Sample T-Test formula.
6. The p-value was calculated to determine whether the difference from the neutral value was statistically significant.

Formula Used

t = (Mean − 3) / (SD / √n)

Example in Excel:

=(3.609090909-3)/(1.181628906/SQRT(110))

Where:

- 3.609090909 = Mean
- 1.181628906 = Standard Deviation
- 110 = Number of respondents

- 3 = Neutral/reference value

Result Obtained

- Mean = 3.609090909
- SD = 1.181628906
- t-value = 5.406265295
- p-value = 3.8317175E-07

Final Value

p-value = 0.000000383172

WORKING SCREENSHOT:

ONE-SAMPLE T-TEST – Q14				
Question Selected				
Q14 – The popularity of cricket limits the growth of other sports in India.				
Test value (hypothesized mean) = 3 (Neutral)				
WHY Q14 WAS SELECTED				
Q14 directly measures the main issue of the research: whether respondents believe cricket's popularity limits the growth of other sports. It uses a 1–5 Likert scale, where 3 represents Neutral. The one-sample t-test checks whether the sample mean is significantly different from 3.				
T-TEST RESULT				
Mean	**SD**	**t-value**	**p-value**	**n**
3.609091	1.181629	5.406265	3.831718E-07	110
FORMULAS USED				
Mean	3.609091			
Standard Deviation	1.181629			
t-value	5.406265			
p-value	3.831718E-07			
Decision	**Since p-value < 0.05, the null hypothesis is rejected.**			

2. Chi-Square Test – Q10 and Q11

Questions Selected

Q10: "Do you think cricket gets more media coverage than other sports?"

Q11: "Do you think this more media coverage affects the popularity of other sports?"

Why were Q10 and Q11 selected?

Q10 was selected because it identifies whether respondents feel that cricket receives more media coverage than other sports.

Q11 was selected because it identifies whether respondents feel that this additional media coverage affects the popularity of other sports.

Both questions are directly connected to the research topic. Therefore, the Chi-Square Test was used to check whether there is a significant association between the two responses.

Steps Followed

1. The responses for Q10 and Q11 were collected from all 110 respondents.
2. The responses were arranged into an observed frequency table.
3. Similar response categories were combined to create the required 2 × 2 contingency table.
4. The observed frequencies were compared with the expected frequencies.
5. The expected frequencies were calculated using the row totals, column totals and grand total.
6. The Chi-Square value was calculated.
7. The p-value was obtained to determine whether a significant association exists between Q10 and Q11.

Formula Used

$\chi^2 = \Sigma\ (\text{Observed} - \text{Expected})^2 / \text{Expected}$

Example in Excel for one cell:

=(Observed_Cell-Expected_Cell)^2/Expected_Cell

For example, if the observed frequency is in B12 and the expected frequency is in C12:

=(B12-C12)^2/C12

Result Obtained

- Chi-Square value = 19.606247
- p-value = 9.51576E-06
- Degrees of freedom = 1
- Significance level = 0.05

Final Value

p-value = 0.00000951576

WORKING SCREENSHOT:

CHI-SQUARE TEST – Q10 AND Q11

Count of 11. Do you think this more	11. Do you think thi…ore media coverage affects the popularity of other sports?			
10. Do you think cricket gets mo…	No	Not Sure	Yes	**Total Result**
Maybe	3	15	9	**27**
No	6	2		**8**
Yes	5	17	53	**75**
Total Result	**14**	**34**	**62**	**110**
Maybe	3	15	9	27
No	6	2	0	8
Yes	**5**	**17**	53	75
Grand Total	14	34	62	110

2 × 2 TABLE USED FOR CHI-SQUARE TEST

Q10 / Q11	No / Not Sure	Yes	Total
No / Maybe	26	9	35
Yes	22	53	75
Total	48	62	110

EXPECTED FREQUENCY

Q10 / Q11	No / Not Sure	Yes
No / Maybe	15.2727	19.7273
Yes	32.7273	42.2727

CHI-SQUARE CONTRIBUTIONS

Q10 / Q11	No / Not Sure	Yes
No / Maybe	7.5346	5.8333
Yes	3.5162	2.7222

CHI-SQUARE TEST RESULT

Chi-Square (χ^2)	19.606247
Degrees of Freedom	1
p-value	9.515757E-06
Significance Level (α)	0.05
Decision	**Since p-value < 0.05, the null hypothesis is rejected.**

3. ANOVA – Q5 and Q14

Questions Selected

Q5: "How often do you watch cricket?"

Q14: "The popularity of cricket limits the growth of other sports in India."

Why were Q5 and Q14 selected?

Q5 was selected because it shows how frequently respondents watch cricket.

The responses were divided into five groups:

- Daily
- Weekly
- Monthly
- Rarely
- Never

Q14 was selected because it measures the respondents' opinion about whether the popularity of cricket limits the growth of other sports.

Therefore, ANOVA was used to determine whether the mean Q14 response differs across the different cricket-watching-frequency groups in Q5.

Steps Followed

1. The responses for Q5 and Q14 were collected from all 110 respondents.
2. The Q14 scores were grouped according to the Q5 categories.
3. Five groups were created: Daily, Weekly, Monthly, Rarely and Never.
4. The number of respondents, sum, average and variance were calculated for each group.
5. The five groups were entered into Excel → Data Analysis → ANOVA: Single Factor.
6. Excel calculated the variation between groups and within groups.
7. The F-value, p-value and F-critical value were obtained from the ANOVA output.

Formula Used

F = MS Between Groups / MS Within Groups

Example in Excel:

=D17/D18

Where:

- D17 = Mean Square Between Groups
- D18 = Mean Square Within Groups

The ANOVA Data Analysis tool in Excel calculates these values automatically.

Result Obtained

- F-value = 1.139661
- p-value = 0.342001
- F-critical = 2.45821

Final Value

p-value = 0.342001

Decision: Since the p-value (0.342001) is greater than 0.05, the null hypothesis is not rejected.

Conclusion: There is no statistically significant difference in the mean Q14 responses among the different cricket-watching-frequency groups.

WORKING SCREENSHOT:

ONE-WAY ANOVA – Q5 AND Q14

Questions Selected

Q5 – How often do you watch cricket?

Q14 – The popularity of cricket limits the growth of other sports in India.

GROUP SUMMARY

Group	Count	Sum	Average	Variance
Daily	10	37	3.700000	0.900000
Weekly	14	51	3.642857	1.478022
Monthly	19	63	3.315789	2.005848
Rarely	50	191	3.820000	1.048571
Never	*17*	*55*	*3.235294*	*1.941176*

ANOVA RESULT

Source of Variation	SS	df	MS	F	p-value	F critical
Between Groups	6.332537	4.000000	1.583134	1.139661	0.342001	2.458210
Within Groups	145.858372	105.000000	1.389127			
Total	152.190909	109.000000				

Decision **Since p-value >= 0.05, the null hypothesis is not rejected.**

4. Correlation – Q14 and Q15

Questions Selected

Q14: "The popularity of cricket limits the growth of other sports in India."

Q15: "Other sports deserve the same level of promotion and media coverage as cricket."

Why were Q14 and Q15 selected?

Q14 was selected because it measures whether respondents feel that cricket's popularity limits the growth of other sports.

Q15 was selected because it measures whether respondents feel that other sports deserve the same level of promotion and media coverage as cricket.

Both questions use a 1–5 Likert scale and are directly related to the main research topic. Therefore, correlation was used to determine whether there is a relationship between the responses to Q14 and Q15.

Steps Followed

1. The numerical responses for Q14 and Q15 were collected from all 110 respondents.
2. The Q14 responses were placed in one column and the Q15 responses in another column.
3. The Pearson correlation coefficient was calculated using the CORREL function in Excel.
4. The correlation coefficient was used to identify the direction and strength of the relationship.
5. The significance of the correlation was tested using the corresponding t-statistic.
6. The p-value was calculated to determine whether the correlation was statistically significant.

Formula Used

Pearson Correlation:

r = CORREL(X,Y)

Example in Excel:

=CORREL(P2:P111,Q2:Q111)

Where:

- P2:P111 = Q14 responses
- Q2:Q111 = Q15 responses

The result gives the Pearson correlation coefficient (r).

Formula for Significance of Correlation

$$t = r\sqrt{(n-2)} / \sqrt{(1-r^2)}$$

Example in Excel:

=C4*SQRT((110-2)/(1-C4^2))

Where:

- C4 = correlation coefficient
- 110 = number of respondents

The p-value was calculated using:

=T.DIST.2T(ABS(C4*SQRT((110-2)/(1-C4^2))),108)

Result Obtained

- Correlation coefficient (r) = 0.546137
- p-value = 6.78E-10
- Degrees of freedom (df) = n − 2 = 110 − 2 = 108

The positive value of **r = 0.546137** indicates a positive relationship between Q14 and Q15.

Final Value

r = 0.546137

p-value = 0.000000000678

Decision: Since the p-value (6.78E-10) is less than 0.05, the null hypothesis is rejected.

Conclusion: There is a statistically significant positive relationship between Q14 and Q15.

WORKING SCREENSHOT:

CORRELATION TEST – Q14 AND Q15

Questions Selected

Q14 – The popularity of cricket limits the growth of other sports in India.
Q15 – Other sports deserve the same level of promotion and media coverage as cricket.

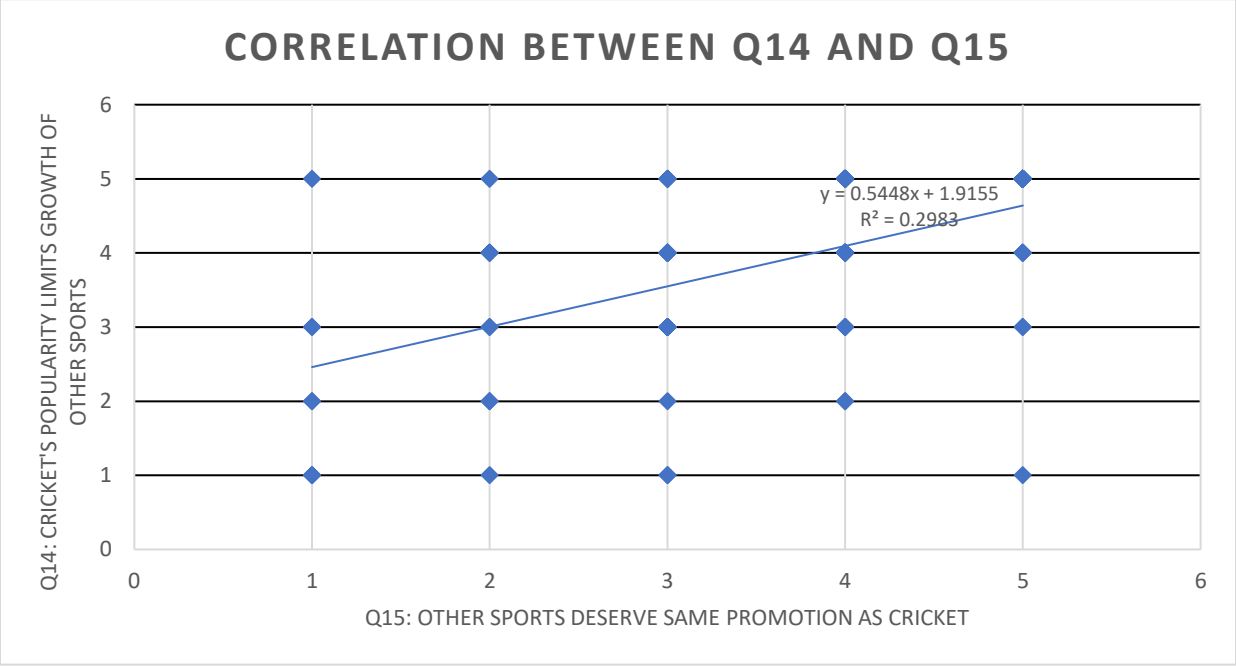
WHY THESE QUESTIONS WERE SELECTED

Q14 measures respondents' view about cricket's impact on the growth of other sports, while Q15 measures their view about equal promotion and media coverage. Correlation tests whether these two responses are related.

CORRELATION RESULT

Measure	Value	Formula
Correlation coefficient (r)	0.546137	0.546136528
p-value	6.782994E-10	6.78299E-10

Decision — **Since p-value < 0.05, the null hypothesis of zero correlation is rejected.**



❖ INTERPRETATION

1. T-Test – Q14

H_0: The mean response for Q14 is not significantly different from the neutral value of 3.

H_1: The mean response for Q14 is significantly different from the neutral value of 3.

The p-value is **3.83172E-07**, which is less than **0.05**.

Decision: Reject H_0.

Interpretation: There is a statistically significant difference between the observed mean response and the neutral value. The mean score of **3.61** shows that respondents generally lean towards agreement with the statement that the popularity of cricket limits the growth of other sports in India.

2. Chi-Square Test – Q10 and Q11

H_0: There is no significant association between the perception that cricket receives more media coverage and the perception that this media coverage affects the popularity of other sports.

H_1: There is a significant association between the two variables.

The p-value is **9.51576E-06**, which is less than **0.05**.

Decision: Reject H_0.

Interpretation: There is a statistically significant association between respondents' views about cricket receiving more media coverage and their views about the effect of this media coverage on the popularity of other sports.

3. ANOVA – Q5 and Q14

H_0: There is no significant difference in the mean Q14 responses among the different cricket-watching-frequency groups.

H_1: There is a significant difference in the mean Q14 responses among the groups.

The p-value is **0.342001**, which is greater than **0.05**.

Decision: Fail to reject H_0.

Interpretation: There is no statistically significant difference in respondents' opinions about cricket limiting the growth of other sports based on how frequently they watch cricket.

4. Correlation – Q14 and Q15

H_0: There is no significant relationship between Q14 and Q15.

H_1: There is a significant relationship between Q14 and Q15.

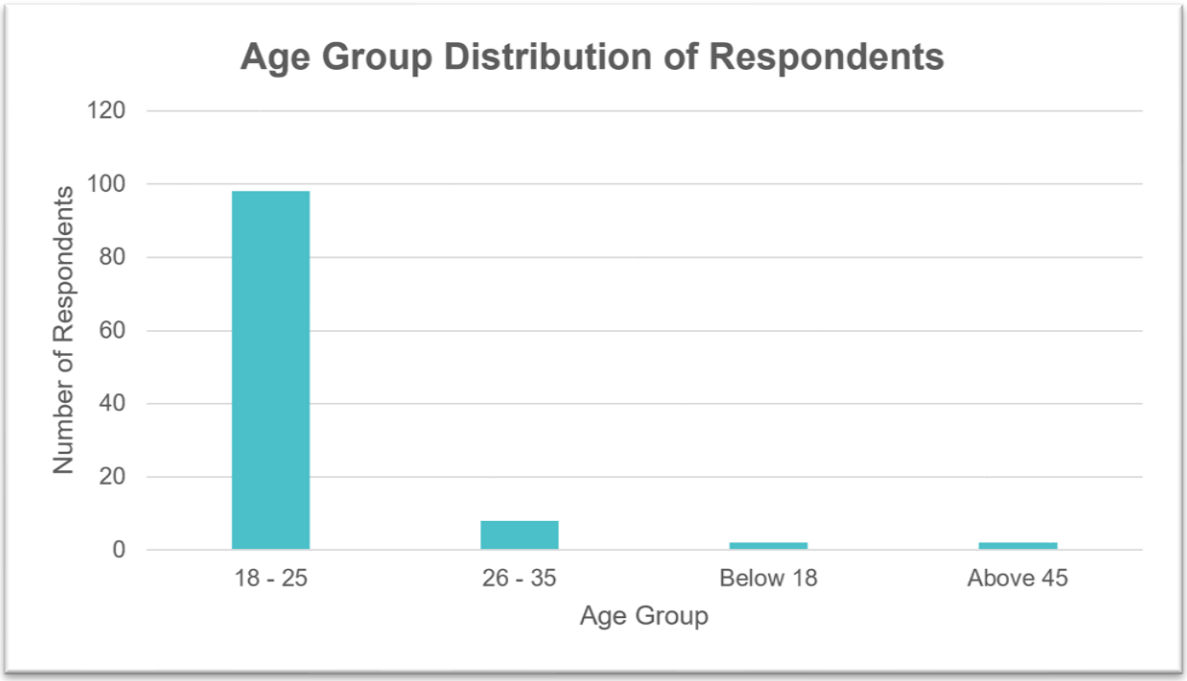
The correlation coefficient is **0.546137** and the p-value is **6.78E-10**, which is less than **0.05**.

Decision: Reject H_0.

Interpretation: There is a statistically significant positive relationship between the two variables. Respondents who tend to agree that cricket's popularity limits the growth of other sports also tend to support giving other sports the same level of promotion and media coverage as cricket.

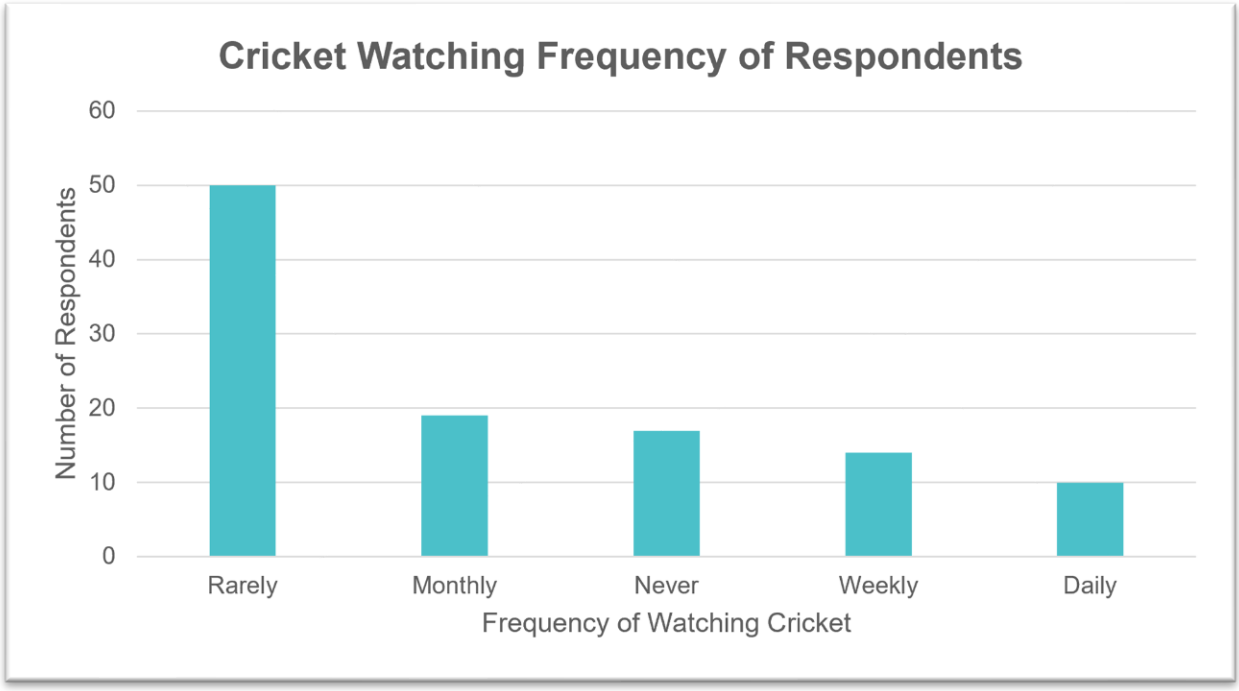
❖ GRAPHS/CHARTS

Chart 1: Age Group Distribution of Respondents



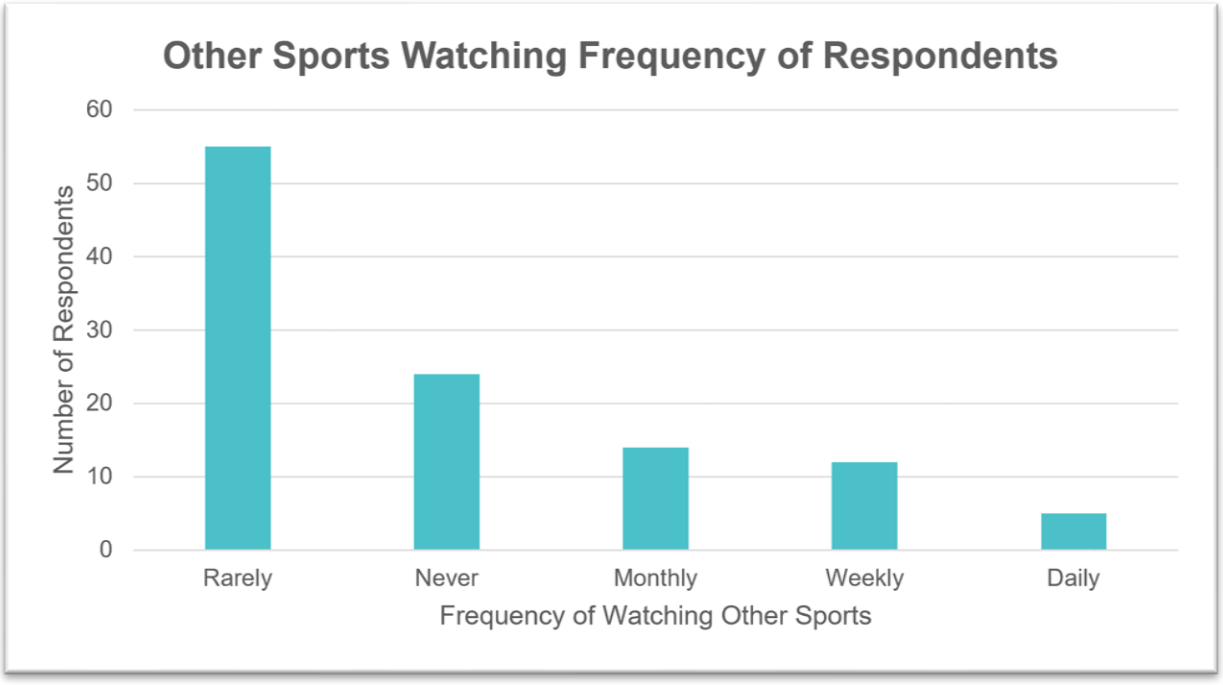
The chart shows that most respondents are from the 18–25 age group (98 respondents). The 26–35 age group has 8 respondents, while Below 18 and Above 45 have 2 respondents each. This indicates that the survey mainly represents young adults aged 18–25.

Chart 2: Cricket Watching Frequency of Respondents



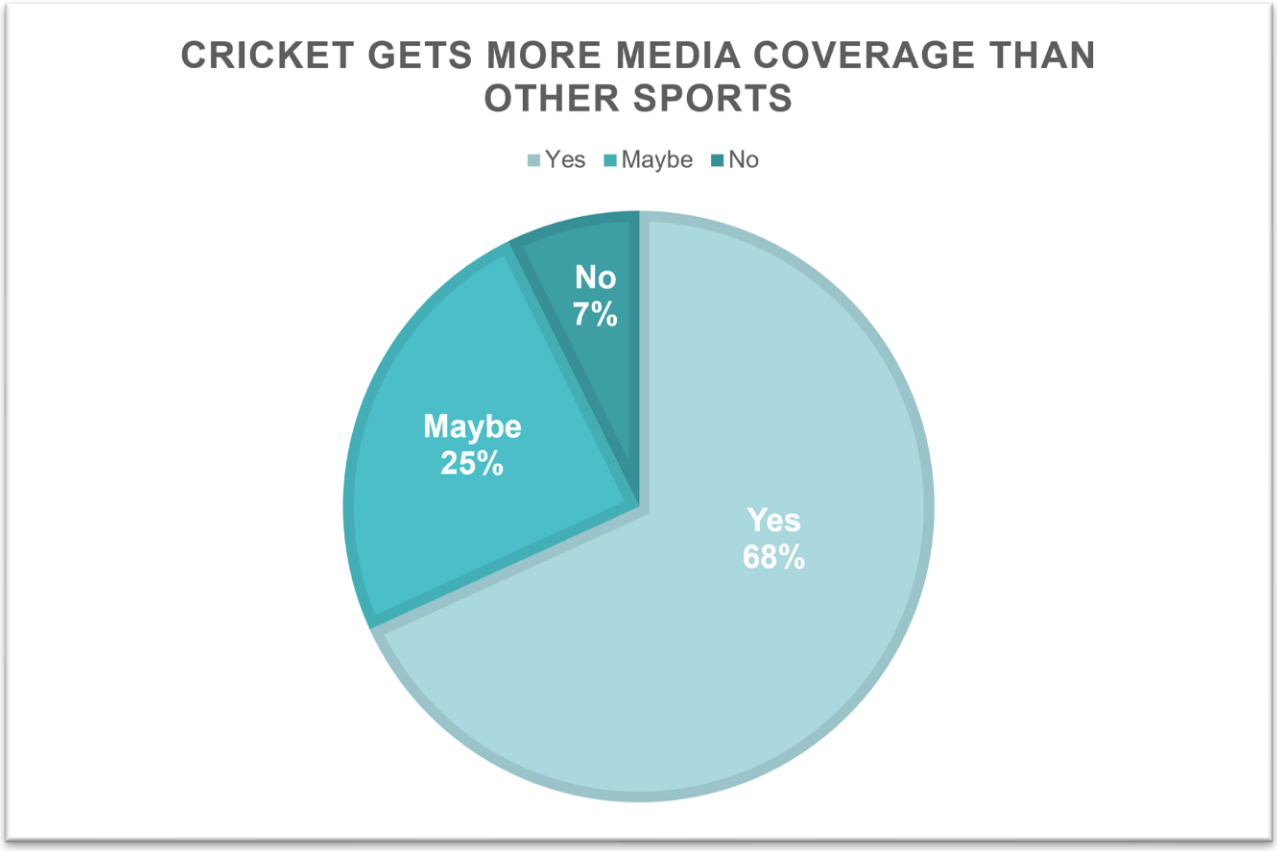
The chart shows the frequency at which respondents watch cricket. 50 respondents watch cricket rarely, making it the most common response. This is followed by 19 respondents who watch monthly, 17 who never watch, 14 who watch weekly, and 10 who watch daily. This indicates that the respondents have varied levels of interest in watching cricket, with rare viewing being the most common.

Chart 3: Other Sports Watching Frequency of Respondents



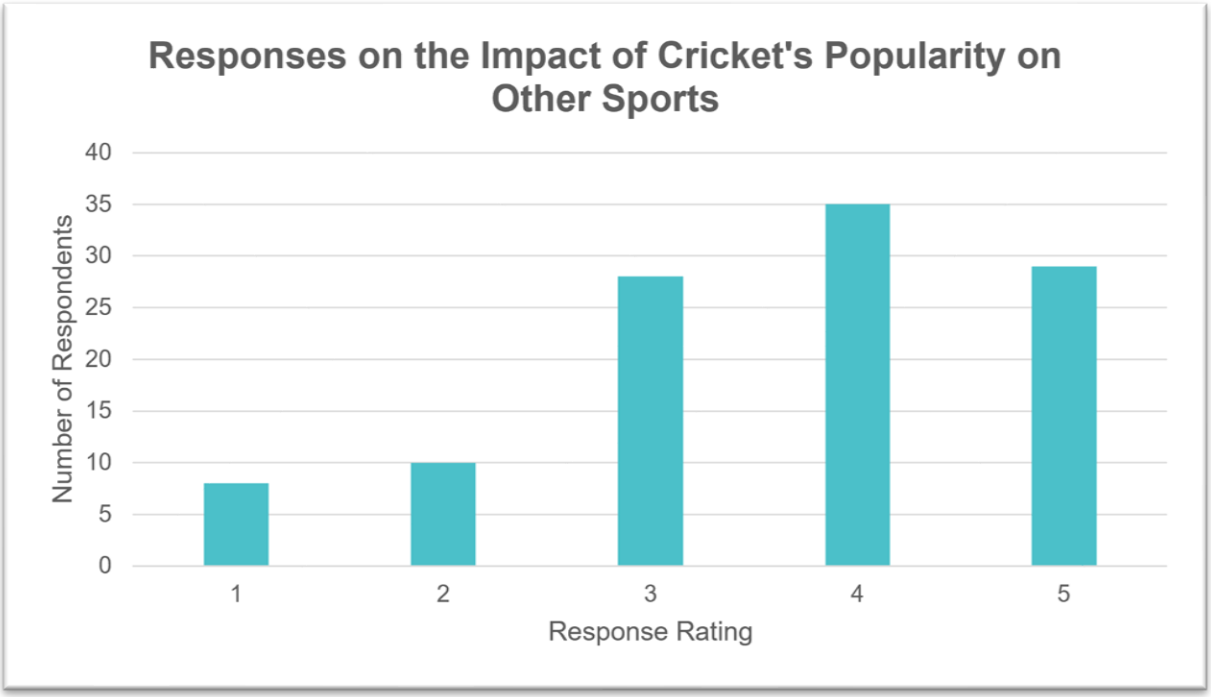
The chart shows how frequently respondents watch sports other than cricket. 55 respondents watch other sports rarely, which is the highest response. This is followed by 24 respondents who never watch, 14 who watch monthly, 12 who watch weekly, and 5 who watch daily. Overall, the chart indicates that respondents watch other sports less frequently, with rare viewing being the most common pattern.

Chart 4: Cricket Gets More Media Coverage Than Other Sports



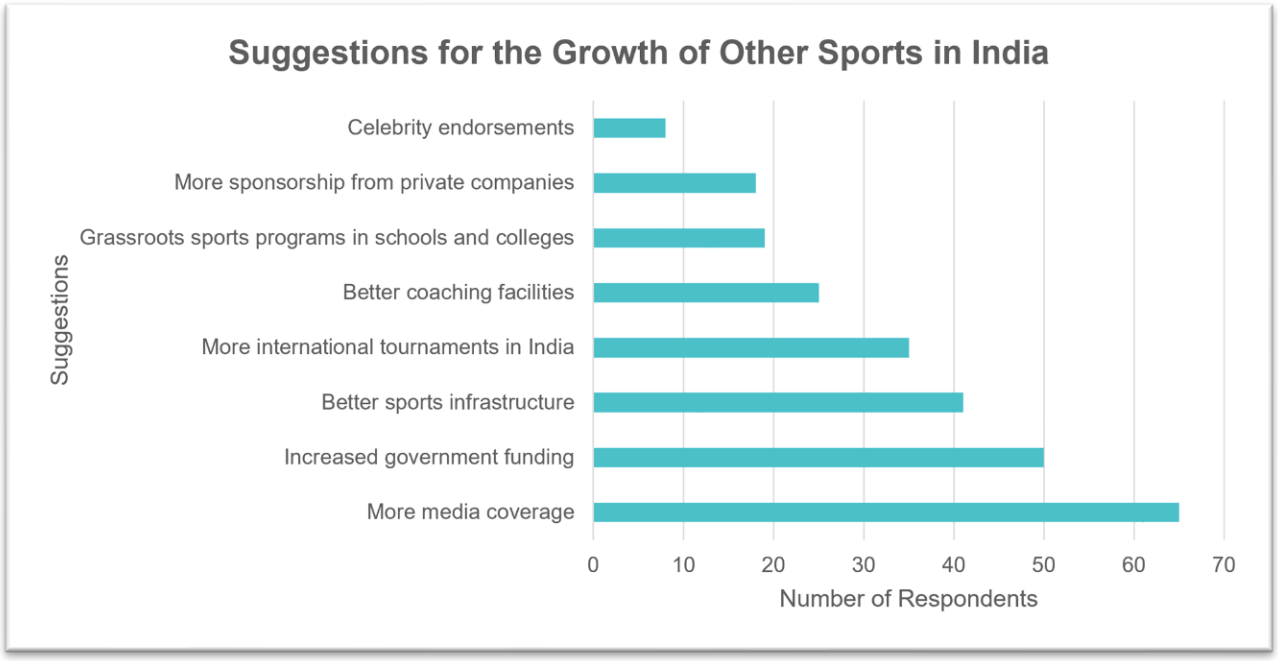
The pie chart shows respondents' opinions on whether cricket gets more media coverage than other sports. 68% (75 respondents) said Yes, while 25% (27 respondents) said Maybe and 7% (8 respondents) said No. This shows that a large majority of respondents believe that cricket receives more media coverage than other sports.

Chart 5: Impact of Cricket's Popularity on Other Sports



The chart shows respondents' ratings on whether the popularity of cricket affects other sports. 35 respondents gave a rating of 4, while 29 gave a rating of 5. 28 respondents rated it 3, followed by 10 respondents with a rating of 2 and 8 with a rating of 1. Overall, most respondents gave higher ratings, indicating that many respondents feel cricket's popularity has an impact on the growth of other sports.

Chart 6: Suggestions for the Growth of Other Sports in India



The chart shows the suggestions given by respondents for promoting the growth of other sports in India. More media coverage received the highest response with 65 respondents, followed by increased government funding (50) and better sports infrastructure (41). Other suggestions included more international tournaments (35), better coaching facilities (25), grassroots sports programs (19), sponsorship from private companies (18), and celebrity endorsements (8). Overall, respondents mainly suggested greater media coverage, funding, and infrastructure to support other sports.

Practical 8:- Writing references

Writing of report is the last step in a research study and requires a set of skills somewhat different from those called for in respect of the earlier stages of research. This task should be accomplished by the researcher with utmost care; he may seek the assistance and guidance of experts for the purpose.

A reference citation is the documentation needed to make your paper acceptable for academic purposes. It gives authoritative sources for your statements, helps the reader gain access to those sources, and acknowledges the fact that the information used in a paper did not originate with the writer

APA style uses the author/date method of citation in which the author's last name and the year of the publication are inserted in the actual text of the paper. It is the style recommended by the American Psychological Association

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REFERENCES:- Books /Websites/research papers /journal articles as per the examples given below:-

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